

Pranav Cavaturu

+1 (469) 350-3101 | pcavaturu00@gmail.com | linkedin.com/in/cavaturu

EDUCATION

The University of Texas at Austin

B.S. in Radio-Television-Film; Certificate in Programming and Computation

Austin, TX

May 2029

EXPERIENCE

Software Engineering Intern

May 2026 – Sep. 2026

XRAY Commerce

Remote

- Built an AI video-generation pipeline turning a single product photo into cinematic ad videos, cutting studio-quality ad production to minutes by orchestrating diffusion models with anti-hallucination prompting
- Improved generated-ad fidelity to real products by engineering a vision-language model verification step that automatically flags and regenerates outputs failing product-accuracy checks
- Increased apparel ad realism by developing an identity-preserving virtual try-on module that defaults garments to on-model shots instead of flat product images

Founder and Full Stack Engineer

Apr. 2026 – Present

Scripy

Austin, TX

- Grew an AI screenwriting platform to 100+ active filmmakers at UT Austin through organic marketing by shipping a full-stack Fountain-format editor that takes a screenplay from concept to production-ready outputs
- Enabled writers to skip \$100-\$400 in professional fees by building an AI script-coverage tool that scores screenplays across 8 studio-standard dimensions and returns industry-format feedback
- Saved production teams 4-8 hours per script revision by developing an engine that auto-generates per-scene production breakdowns on every screenplay edit

Software Developer

Sep. 2025 – Present

Longhorn Developers

Austin, TX

- Shipped features to production on a 15+ engineer Agile team, building React/TypeScript components and merging pull requests through CI/CD

PROJECTS

Kubrick | *Python, FastAPI, PyTorch, Modal, Transformers, React, TypeScript*

2026

- Enabled filmmakers to score trailers for predicted audience attention by engineering and deploying a system that forecasts per-shot viewer neural response to video, built on top of the Tribe V2 model
- Powered model inference by building an end-to-end pipeline that segments video with ffmpeg, transcribes speech with WhisperX, and extracts audio features for feature-level prediction
- Cut infrastructure overhead by deploying the model as a serverless endpoint on Modal, securing webhooks with HMAC authentication to sanitize incoming jobs and persist outputs

B3VO AI | *React Native, Expo, TypeScript, ElevenLabs, Twilio*

2026

- Delivered an anonymous AI mental-health companion to UT Austin students over SMS and iMessage by fine-tuning conversational flows on Cognitive Behavioral Therapy and mindfulness techniques
- Drove sustained engagement by building a native iOS app with daily habit tracking (sleep, nutrition, mood, journaling) and streak logic, plus guided breathing and ElevenLabs-narrated body scans
- Guaranteed user privacy by architecting a local-first app with anonymous device identity and no cloud dependency

CAMPUS | *React Native, TypeScript, Node.js*

2025

- Unified fragmented campus tools into one product by building a social-discovery platform combining event discovery, fundraisers, a peer marketplace, and gamified campus quests
- Increased content engagement by designing and building Buzz, a campus-scoped anonymous short-form video feed that lets posts embed a linked event, fundraiser, or marketplace listing

TECHNICAL SKILLS

Languages: Python, Java, C/C++, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks: React, React Native, Expo, Next.js, Node.js, FastAPI, PyTorch, Transformers, Tailwind CSS, Three.js

Cloud & Infra: AWS (S3), Cloudflare R2, Modal, Docker, CI/CD, Convex

Developer Tools: Git, Stripe, Clerk, REST APIs

Concepts: Data Structures & Algorithms, Distributed Systems, Unit Testing, Vision-Language Models, LLM Prompting